

PHARMACOLOGY

Antidepressant drugs

1. Overview

The symptoms of depression are feelings of sadness and hopelessness as well as the inability to experience pleasure in usual activities changes in sleep patterns and appetite, loss of energy, and suicidal thoughts.

2. General mechanism of antidepressant drugs

Most clinically useful antidepressant drugs potentiate, either directly or indirectly, the actions of norepinephrine and/or serotonin (5-HT) in the brain. This led to the biogenic amine theory, which proposes that depression is due to a deficiency of monoamines, such as norepinephrine and serotonin, at certain key sites in the brain.

3. Classes of antidepressant drugs

- 1-Selective serotonin reuptake inhibitors (SSRIs)
- 2- Serotonin/norepinephrine reuptake inhibitors (SNRIs)
- 3- Tricyclic antidepressants (TCAs)
- 4- Monoamine oxidase inhibitors (MAOIs)
- 5- Atypical antidepressants

1- Selective serotonin reuptake inhibitors (SSRIs)

- The selective serotonin reuptake inhibitors (SSRIs) are a group of antidepressant drugs that specifically inhibit serotonin reuptake
- SSRIs have little blocking activity at muscarinic, α -adrenergic, and histaminic H₁ receptors.

Drug examples

Fluoxetine (the prototypic drug), **citalopram**, **escitalopram**, is the pure S-enantiomer of citalopram, **fluvoxamine**, **paroxetine**, and **sertraline**.

Mechanism of action

- The SSRIs block the reuptake of serotonin, leading to increased concentrations of the neurotransmitter in the synaptic cleft.
- SSRIs, typically take at least 2 weeks to produce significant improvement in mood, and maximum benefit may require up to 12 weeks or more.

Therapeutic uses

- Depression
- Obsessive–compulsive disorder, panic disorder, generalized anxiety disorder, posttraumatic stress disorder, social anxiety disorder, premenstrual dysphoric disorder
- Bulimia nervosa (only fluoxetine is approved for bulimia).

Pharmacokinetics

- All of the SSRIs are well absorbed after oral administration.
- Peak levels are seen in approximately 2 to 8 hours.
- Food has little effect on absorption (except with sertraline, for which food increases its absorption).
- The majority of SSRIs have plasma half-lives that range between **16 and 36** hours.
- Metabolism by cytochrome P450 (CYP450)–dependent enzymes and glucuronide or sulfate conjugation occur extensively.
- Fluoxetine differs from the other members of the class by having a much longer half-life (50 hours)
- Fluoxetine and paroxetine are potent inhibitors of a CYP450 isoenzyme (CYP2D6)

- Dosages of the SSRIs should be reduced in patients with hepatic impairment.

Adverse effects

- 1- **GIT adverse effects**
- 2- **Sleep disturbances**
- 3- **Sexual dysfunction:** may include loss of libido, delayed ejaculation, is common with the SSRIs.
- 4- **Used cautiously in children and teenagers**
- 5- **SSRI does not usually cause cardiac arrhythmias except** with citalopram, which may cause QT prolongation.
- 6- **Seizures**
- 7- **Serotonin syndrome**
- 8- **Discontinuation syndrome**

2- Serotonin/norepinephrine reuptake inhibitors (SNRIs)

Drug examples

Venlafaxine, desvenlafaxine, levomilnacipran, and duloxetine

Mechanism of action

They inhibit the reuptake of both serotonin and norepinephrine

Therapeutic uses

- Effective in treating depression in patients in whom SSRIs are ineffective.
- In relieving pain associated with diabetic peripheral neuropathy, postherpetic neuralgia, fibromyalgia, and low back pain.

A. Venlafaxine and desvenlafaxine

- Venlafaxine is a potent inhibitor of serotonin reuptake and, at medium to higher doses, is an inhibitor of norepinephrine reuptake.

- Venlafaxine has minimal inhibition of the CYP450 isoenzymes and is a substrate of the CYP2D6 isoenzyme.
- **Side effects** of venlafaxine are GIT disturbances with troublesome nausea, sexual dysfunction, dizziness, and insomnia
- At high doses, there may be an increase in blood pressure and heart rate. The clinical activity and adverse effect profile of desvenlafaxine are similar to that of venlafaxine.

B. Duloxetine

- Duloxetine inhibits serotonin and norepinephrine reuptake at all doses.
- It is extensively metabolized in the liver to inactive metabolites
- GIT side effects are common with duloxetine; including nausea, dry mouth, and constipation. Sleep disturbances, and sexual dysfunction are also seen.
- Duloxetine may increase blood pressure or heart rate.
- Duloxetine is a moderate inhibitor of CYP2D6 isoenzymes and may increase concentrations of drugs metabolized by this pathway, such as antipsychotics.

C. Levomilnacipran

- The adverse effect profile of *levomilnacipran* is similar to other SNRIs.
- It is primarily metabolized by CYP3A4, and, thus, activity may be altered by inducers or inhibitors of this enzyme system

3- Tricyclic antidepressants (TCAs)

Drug examples

Tertiary amines: imipramine (the prototype drug), amitriptyline, clomipramine, doxepin, and trimipramine.

Secondary amines: desipramine and nortriptyline (the N-demethylated metabolites of imipramine and amitriptyline, respectively) and protriptyline.

Maprotiline and amoxapine are related “tetracyclic” antidepressant agents and are commonly included in the general class of TCAs

Mechanism of action

- 1- **Inhibition of neurotransmitter reuptake:** TCAs and amoxapine are potent inhibitors of the neuronal reuptake of norepinephrine and serotonin into presynaptic nerve terminals. Maprotiline and desipramine are relatively selective inhibitors of norepinephrine reuptake=
- 2- **Blocking of other receptors:** TCAs also block serotonergic, α -adrenergic, histaminic, and muscarinic receptors which actions at these receptors are likely responsible for many of their adverse effects.

Therapeutic uses

- Moderate to severe depression.
- Panic disorder.
- Prevent migraine headache and treat chronic pain syndromes (for example, neuropathic pain) , particularly amitriptyline
- Low doses of TCAs, especially doxepin, can be used to treat insomnia.
- Nocturnal enuresis.

Pharmacokinetics

- TCAs are well absorbed upon oral administration.
- Because of their lipophilic nature, they are widely distributed and readily penetrate into the CNS.
- TCAs have low and inconsistent bioavailability.
- Metabolized by the hepatic microsomal system and excreted as inactive metabolites via the kidney.

Adverse effects

- Blockade of muscarinic receptors leads to blurred vision, xerostomia dry mouth), urinary retention, sinus tachycardia, constipation, and aggravation of angle-closure glaucoma
- May precipitate life-threatening arrhythmias in an overdose.
- Blockade of α -adrenergic receptors, causing orthostatic hypotension, dizziness, and reflex tachycardia.
- Blockade of histamine H1 receptors leads to sedation
- Weight gain is a common adverse effect of the TCAs.
- Sexual dysfunction occurs in a minority of patients
- TCAs (like all antidepressants) should be used with caution in patients with bipolar disorder, even during their depressed state
- The TCAs have a narrow therapeutic index

4. Monoamine oxidase inhibitors

- The MAOIs may irreversibly or reversibly inactivate the enzyme MAO, permitting neurotransmitters to escape degradation and, therefore, to accumulate within the presynaptic neuron and leak into the synaptic space.
- **Drugs:** phenelzine, tranylcypromine, isocarboxazid and selegiline

Mechanism of action

- Most MAOIs, such as phenelzine, form stable complexes with the enzyme, causing irreversible inactivation. This results in increased stores of norepinephrine, serotonin, and dopamine within the neuron and subsequent diffusion of excess neurotransmitter into the synaptic
- These drugs inhibit not only MAO in the brain but also MAO in the liver and gut

Therapeutic uses

- In depressed patients who are unresponsive or allergic to TCAs and SSRIs or who experience strong anxiety.
- Atypical depression.
- Because of their risk for drug– drug and drug–food interactions, the MAOIs are considered last-line agents in many treatment settings.

Pharmacokinetics

- These drugs are well absorbed after oral administration.
- MAOIs are hepatically metabolized and excreted rapidly in urine.

Adverse effects

- Due to drug–food and drug–drug interactions, limit the widespread use of MAOIs. For example, a hypertensive crisis with tyramine, which is contained in foods, such as aged cheeses and meats, chicken liver, pickled or smoked fish, and red wines, is normally inactivated by MAO in the gut.
- Drowsiness, orthostatic hypotension, blurred vision, dry mouth, and constipation.
- Serotonin syndrome when used with other antidepressants

5. Atypical antidepressants

The atypical antidepressants are a mixed group of agents that have actions at several different sites.

Drugs: bupropion, mirtazapine, nefazodone, trazodone,

A. Bupropion

- Bupropion is a weak dopamine and norepinephrine reuptake inhibitor
- Bupropion is also useful for decreasing cravings and attenuating withdrawal symptoms of nicotine in patients trying to quit smoking.
- **Side effects** may include dry mouth, sweating, nervousness, tremor, and a dose dependent increased risk for seizures.
- It has a very low incidence of sexual dysfunction.

- Bupropion is metabolized by the CYP2B6 pathway and has a relatively low risk for drug–drug interactions

B. Mirtazapine

- Mirtazapine enhances serotonin and norepinephrine neurotransmission by serving as an antagonist at presynaptic α_2 receptors.
- Increased appetite and weight gain frequently occur
- Mirtazapine is markedly sedating, which may be an advantage in depressed patients having difficulty sleeping.
- It has a very low incidence of sexual dysfunction

C. Nefazodone and trazodone

- These drugs are weak inhibitors of serotonin reuptake.
- Their therapeutic benefit appears to be related to their ability to block postsynaptic 5HT_{2A} receptors.
- Both agents are sedating
- Trazodone has been associated with priapism, and nefazodone has been associated with a risk for hepatotoxicity.
- Both agents also have mild to moderate α_1 receptor antagonism, contributing to orthostasis and dizziness=

D. Vilazodone

- Vilazodone is a serotonin reuptake inhibitor and a 5-HT_{1a} partial agonist. The adverse effect profile of vilazodone is similar to the SSRIs, including a risk for discontinuation syndrome if abruptly stopped=

E. Vortioxetine

- Vortioxetine utilizes a combination of serotonin reuptake inhibition, 5HT_{1a} agonism, and 5-HT₃ and 5-HT₇ antagonism.
- The common adverse effects include nausea, vomiting, and constipation.

New antidepressant medications: See the lecture